

ORIGINAL ARTICLE

# Harms are assessed inconsistently and reported inadequately Part 2: nonsystematic adverse events

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## Abstract

**Objective:** We examined nonsystematic adverse events (AEs) in Part 2 (of 2) of a study describing the assessment and reporting AEs in clinical trials.

**Study Design and Setting:** We examined 21 trials of gabapentin for neuropathic pain (52 sources) and seven trials of quetiapine for bipolar depression (80 sources) using data from the Multiple Data Sources study. We extracted and compared information about nonsystematic AEs (i.e., AEs that were not assessed for every participant), including AEs categorized as “serious.” We recorded whether AEs were grouped by anatomic or physiological system.

**Results:** Trials of the same drug reported information about different AEs. Information in public sources was inadequate for decision-making. No public source reported all AEs, or all serious AEs, identified in nonpublic sources about the same trial. Of trials with only public sources, 2/15 (13%) gabapentin and 0/3 (0%) quetiapine trials grouped AEs by anatomic or physiological system.

**Conclusion:** Public sources contained little information about nonsystematic AEs, including serious AEs. Grouping might make nonsystematic AEs easier to detect; however, most public sources did not report grouped AEs. Standards are needed to improve the collection and reporting of nonsystematic AEs so that stakeholders can use trials to assess the balance of potential benefits and harms. © 2019 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

**Keywords:** Adverse events; Clinical trials; Drug safety; Reporting bias; Open science; Harms

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### What is new?

- The presence or absence of a nonsystematic adverse event (AEs) is not assessed for all participants in a clinical trial; instead, nonsystematic AEs are reported to investigators in response to open-ended questions such as “have you noticed any symptoms since your last visit?”
- Trials of the same intervention for the same condition include information about different nonsystematic AEs.
- Most nonsystematic AEs, including serious AEs, were not reported in public sources; this suggests that nonsystematic AEs might be “cherry-picked” for reporting.
- Nonsystematic AEs can be grouped for analysis (e.g., to increase statistical power), but few public sources reported grouped AEs.
- Nonsystematic assessment leads to numerous problems for interpretation and synthesis. All AEs that occur in clinical trials should be reported publicly (e.g., on [www.ClinicalTrials.gov](http://www.ClinicalTrials.gov)).

## 1. Introduction

It is difficult for patients and clinicians to compare the adverse events (AEs) associated with different interventions because AEs are not reported completely in public sources, such as journal articles [1–7]. Nonpublic sources, such as clinical study reports (CSRs), might contain more information [8–11]. Furthermore, multiple sources about clinical trials may contain different information about AEs [8–15].

The methods used to assess and report AEs contribute to challenges for interpretation and synthesis, which previous guidelines have sought to address [16]. Although AEs can be assessed systematically as described in Part 1 of this study [17]—using methods such as those used to assess potential benefits [18,19]—AEs can also be assessed nonsystematically (Table 1) [20,21]. Nonsystematic AEs may be reported by participants to investigators or collected in response to open-ended questions such as “have you noticed any symptoms since your last visit?” AEs may be categorized as “serious” when they lead to hospitalization or death or disrupt normal life functions [22–24].

Terms used to describe AEs are often related (e.g., “migraine” and “headache”). Because many AEs are rare, grouping AEs by anatomic or physiological system can increase statistical power and thus increase the possibility of detecting AEs that are caused by interventions. Grouping AEs can also disguise important AEs by combining them with less important AEs (e.g., “migraine” might be more severe than “headache”). When information about

nonsystematic AEs is submitted to regulators, AEs are often reported in groups (e.g., “nervous system disorders”). For submissions to the U.S. Food and Drug Administration (FDA), the Medical Dictionary for Regulatory Activities (MedDRA) [25] classification system replaced COSTART [26] in the late 1990s.

Part 1 and Part 2 of this study about AEs are substudies of the Multiple Data Sources (MUDS) study, which followed a published protocol [27] and protocol amendments [19]. In this substudy, our objectives were to (1) describe the assessment and reporting of nonsystematic AEs identified in eligible clinical trials and (2) compare reporting of AEs in public and nonpublic sources. We assessed nonsystematic AEs (Part 2) apart from systematic AEs (Part 1) because differences in data collection and analysis might contribute to differences in reporting.

## 2. Methods

### 2.1. Eligible trials and sources

Eligible studies were parallel, randomized controlled trials that evaluated either gabapentin for neuropathic pain or quetiapine for bipolar depression, compared with placebo. We selected these case studies because we had access to both public and nonpublic sources for some of the eligible trials. We searched for public and nonpublic sources (see Table 1) and requested additional information from manufacturers of gabapentin and quetiapine [31]. We included 21 eligible gabapentin trials (80 sources, including six individual patient datasets [IPD]) and seven eligible quetiapine trials (52 sources, including two IPD).

### 2.2. Data extraction

From each source of each eligible trial, two investigators independently extracted data using the open access Systematic Review Data Repository (<http://sdr.ahrq.gov/>) and resolved differences by discussion. We classified each AE as “systematic” if its presence or absence was recorded for every participant and assessed using specific measurement tools (e.g., questionnaires, checklists, laboratory tests, or clinical examinations); we classified all other AEs as “nonsystematic” (Table 1).

From each source for each trial, we extracted the number of participants who experienced:

- (1) One or more nonsystematic AEs;
- (2) One or more *serious* nonsystematic AEs (i.e., events that were categorized as “serious” by the trial investigators);
- (3) Each nonsystematic AE, which we recorded as the name of the AE (e.g., “dizziness,” “headache”);
- (4) Each serious nonsystematic AE; and

- (5) Each group of AEs (i.e., classified according to anatomic or physiological categories by the trial investigators).

For each AE, we extracted numerical results closest to 8 and 18 weeks. We extracted results that were reported separately for each trial; we did not extract results of analyses that pooled multiple trials.

For each reported result, we assessed whether we could estimate the size of the difference between intervention arms and the precision of that estimate (Table 1). All counts of AEs and results are “unique,” meaning that each AE was counted only once, regardless of how many times it appeared across sources (Table 1).

### 2.3. Reanalysis of IPD

For gabapentin, IPD were available for 6 of 21 (29%) trials [32–37] as Microsoft Access databases. For two quetiapine trials [38,39], we used ABBYY FineReader [40] to extract IPD from tables in an appendix of a CSR that we received as a PDF file; we constructed an electronic database for analysis. Because we did not receive meta-data (e.g., codebooks) for the IPD in either case, we used information in the corresponding CSRs (including incomplete case report forms) to determine the meaning of the variables in the IPD. When both IPD and a CSR were available for a trial, we analyzed the IPD using the methods for handling missing data described in the corresponding CSR, and we compared our results with the CSR to assess agreement.

### 2.4. Comparison of reporting across multiple data sources

To determine whether *any* AEs were reported in public and nonpublic sources, we compared:

- (1) the proportion of public and nonpublic sources that reported nonsystematic AEs; and
- (2) the proportion of public and nonpublic sources that reported *serious* nonsystematic AEs.

To determine whether *all* AEs were reported in public and nonpublic sources, we compared:

- (1) the proportion of nonsystematic AEs reported in public and nonpublic sources, including the proportion of nonsystematic AEs for which we could calculate the between-arm effect and its precision;
- (2) The proportion of *serious* nonsystematic AEs reported in public and nonpublic sources, including the proportion of nonsystematic AEs for which we could calculate the between-arm effect and its precision.
- (3) The mean number of AEs reported in public and nonpublic sources; and

- (4) The proportion of trials that reported the five most common and 10 randomly selected AEs (Table 2).

To determine whether “grouped” AEs were reported in public and nonpublic sources, we compared:

- (1) The proportion of public and nonpublic sources that reported AEs for each anatomic or physiological group.

To determine whether public and nonpublic sources included sufficient information about the results, we compared:

- (1) the proportion of trials that reported results for each AE that were sufficient to calculate the between-arm effect and its precision (Table 3, Appendix 1).

We performed all analyses using Stata 14 [41].

## 3. Results

### 3.1. Public sources reported AEs inconsistently and inadequately

We found one or more public sources for 20 of 21 (95%) gabapentin trials and seven of seven (100%) quetiapine trials. Most public sources did not report the number of participants who experienced one or more AEs (often described as “any adverse event”); 5 of 20 (25%) gabapentin trials and three of seven (43%) quetiapine trials reported the number of participants who experienced one or more AEs with sufficient information to calculate the between-arm effect and its precision.

By contrast, we found one or more nonpublic sources for 6 of 21 (29%) gabapentin trials and four of seven (57%) quetiapine trials, and all (100%) nonpublic sources reported the number of participants who experienced one or more AEs with sufficient information to calculate the between-arm effect and its precision.

We found both public and nonpublic sources for 5 of 21 (24%) gabapentin trials and four of seven quetiapine (57%) trials. For trials with both public and nonpublic sources, public sources reported few of the AEs we found in nonpublic sources (Fig. 1).

Across all sources for all trials, we found 419 and 471 unique AEs for gabapentin and quetiapine, respectively. Public sources did not report the vast majority of them; 341 of 419 (81%) and 436 of 471 (93%) unique AEs were *not* reported in public sources (Appendix 1). Public sources reported fewer AEs than nonpublic sources for gabapentin (mean [standard deviation] = 3 [6] public, 121 [40] nonpublic) and quetiapine (three [5] public, 159 [158] nonpublic).

Common AEs were relatively likely to appear in public sources, but other AEs were unlikely to appear in public sources (Table 2, Appendix 1). Without access to nonpublic sources, it was unclear whether AEs that were not mentioned in public sources did not occur or whether those AEs occurred and were not reported.

**Table 1.** Glossary of terms related to adverse events and sources

| Term                                                               | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terms related to adverse events (AEs)</b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Adverse event (AE)                                                 | The International Conference on Harmonisation (ICH) defines an “adverse event” as “any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment” [24]. The U.S. Food and Drug Administration (FDA) and other regulators use this definition [22,23].                                                                                                                                                                                             |
| Coding Symbols for a Thesaurus of Adverse Reaction Terms (COSTART) | “The Coding Symbols for a Thesaurus of Adverse Reaction Terms (COSTART) is the terminology developed and used by the Food and Drug Administration for the coding, filing and retrieving of post-marketing adverse reaction reports. It provides a method to deal with the variation in vocabulary used by those submit adverse event reports to the FDA” [26].                                                                                                                                                                                                                                 |
| Medical Dictionary for Regulatory Activities (MedDRA)              | “The Medical Dictionary for Regulatory Activities (MedDRA) Terminology is the international medical terminology developed under the auspices of the International Conference on Harmonisation (ICH) of Technical Requirements for Registration of Pharmaceuticals for Human Use” [25]. MedDRA replaced COSTART in the late 1990s [28].                                                                                                                                                                                                                                                         |
| Non-systematic adverse events                                      | According to The Final Rule [20,21], “‘non-systematic assessment’ relies on the spontaneous reporting of adverse events, such as unprompted self-reporting by participants.” Non-systematic adverse events may be collected by asking questions such as “Have you noticed any symptoms since your last examination?”                                                                                                                                                                                                                                                                           |
| Result                                                             | In the Multiple Data Source (MUDS) study, a “result” is a numerical contrast between a treatment and comparison arm (e.g., relative risk, mean difference).                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Serious adverse events                                             | The International Conference on Harmonisation (ICH) defines a “serious adverse event” as that which “results in death, is life-threatening, requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability/incapacity, or is a congenital anomaly/birth defect” [24]. FDA and other regulators use this definition [22,23].                                                                                                                                                                                                   |
| Systematic adverse events                                          | According to The Final Rule [20,21], “‘systematic assessment’ involves the use of a specific method of ascertaining the presence of an adverse event (e.g., the use of checklists, questionnaires, specific laboratory tests at regular intervals)”. Like a potential benefit of treatment, a systematic AE can be defined using five elements: (1) domain, (2) specific measurement, (3) specific metric, (4) method of aggregation, and (5) time-point [29]. For example, “proportion of participants with 50% change from baseline to 8 weeks on the Young Mania Rating Scale total score.” |
| Unique adverse event                                               | In the MUDS study, a “unique” AE is a defined AE counted only once, regardless of how many times it appeared in all sources for a trial.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Terms related to sources</b>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Clinical Study Report (CSR)                                        | A comprehensive document, often created by a pharmaceutical manufacturer for submission to a regulator, detailing the design, methods, analyses, and results of a study. Appendices sometimes contain tables of individual patient data, also called “patient data listings,” and study protocols [30].                                                                                                                                                                                                                                                                                        |
| Clinical study report synopsis (CSR-synopsis)                      | A document that summarizes the information contained in a clinical study report. Clinical study report-synopses are much shorter than clinical study reports; the two clinical study report-synopses we examined were each 13 pages in length.                                                                                                                                                                                                                                                                                                                                                 |
| Individual patient data (IPD)                                      | A table or database in which each record contains data for a single participant [30].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Non-public sources                                                 | In the MUDS study, non-public sources include individual patient data, clinical study reports, and clinical study report-synopses.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Public sources                                                     | In the MUDS study, public sources include journal articles, conference abstracts, commentaries, posters, trial registrations and associated results, and Medical Reviews and Statistical Reviews written by the FDA.                                                                                                                                                                                                                                                                                                                                                                           |

### 3.2. Public sources reported serious AEs inconsistently and inadequately

Among trials with public sources, 4 of 20 (20%) gabapentin trials and four of seven (57%) quetiapine trials reported the number of participants who experienced one or more serious AEs with sufficient information to calculate the between-arm effect and its precision. Among trials with

nonpublic sources, six of six (100%) gabapentin trials and four of four (100%) quetiapine trials reported the number of participants who experienced one or more serious AEs with sufficient information to calculate the between-arm effect and its precision.

Across all sources for eligible trials, we found 72 and 46 unique serious AEs in gabapentin and quetiapine

**Table 2.** Public sources describe common adverse events rather than all adverse events.

|                                          | Number of trials with results that include sufficient information to calculate a between-arm effect and its precision ( <i>n</i> ) |                                                          |                                                     |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------|
|                                          | Public sources<br>( <i>N</i> = 20 trials) <i>n</i> (%)                                                                             | Nonpublic sources<br>( <i>N</i> = 6 trials) <i>n</i> (%) | All sources<br>( <i>N</i> = 21 trials) <i>n</i> (%) |
| <b>Gabapentin for neuropathic pain</b>   |                                                                                                                                    |                                                          |                                                     |
| Five most common AEs                     |                                                                                                                                    |                                                          |                                                     |
| Dizziness                                | 9 (45)                                                                                                                             | 6 (100)                                                  | 11 (52)                                             |
| Edema peripheral                         | 5 (25)                                                                                                                             | 6 (100)                                                  | 9 (43)                                              |
| Headache                                 | 8 (40)                                                                                                                             | 6 (100)                                                  | 12 (57)                                             |
| Nausea                                   | 8 (40)                                                                                                                             | 6 (100)                                                  | 12 (57)                                             |
| Somnolence                               | 9 (45)                                                                                                                             | 6 (100)                                                  | 11 (52)                                             |
| Ten randomly selected AEs                |                                                                                                                                    |                                                          |                                                     |
| Abdominal pain                           | 1 (5)                                                                                                                              | 6 (100)                                                  | 6 (29)                                              |
| Anemia                                   | 0 (0)                                                                                                                              | 3 (50)                                                   | 3 (14)                                              |
| Chest pain                               | 0 (0)                                                                                                                              | 5 (83)                                                   | 5 (24)                                              |
| Constipation                             | 1 (5)                                                                                                                              | 6 (100)                                                  | 7 (33)                                              |
| Hernia                                   | 0 (0)                                                                                                                              | 2 (33)                                                   | 2 (10)                                              |
| Hyperglycemia                            | 0 (0)                                                                                                                              | 6 (100)                                                  | 6 (29)                                              |
| Migraine                                 | 0 (0)                                                                                                                              | 3 (50)                                                   | 3 (14)                                              |
| Skin ulcer                               | 0 (0)                                                                                                                              | 3 (50)                                                   | 3 (14)                                              |
| Tooth ache                               | 0 (0)                                                                                                                              | 1 (17)                                                   | 1 (5)                                               |
| Upper respiratory tract infection        | 1 (5)                                                                                                                              | 1 (17)                                                   | 2 (10)                                              |
| <b>Quetiapine for bipolar depression</b> |                                                                                                                                    |                                                          |                                                     |
|                                          | Number of trials with results that include sufficient information to calculate a between-arm effect and its precision              |                                                          |                                                     |
|                                          | Public sources<br>( <i>N</i> = 7 trials) <i>n</i> (%)                                                                              | Nonpublic sources<br>( <i>N</i> = 4 trials) <i>n</i> (%) | All sources<br>( <i>N</i> = 7 trials) <i>n</i> (%)  |
| Five most common AEs                     |                                                                                                                                    |                                                          |                                                     |
| Dizziness                                | 6 (86)                                                                                                                             | 4 (100)                                                  | 6 (86)                                              |
| Dry mouth                                | 6 (86)                                                                                                                             | 4 (100)                                                  | 6 (86)                                              |
| Headache                                 | 6 (86)                                                                                                                             | 4 (100)                                                  | 6 (86)                                              |
| Sedation                                 | 5 (71)                                                                                                                             | 4 (100)                                                  | 5 (71)                                              |
| Somnolence                               | 6 (86)                                                                                                                             | 4 (100)                                                  | 6 (86)                                              |
| Ten randomly selected AEs                |                                                                                                                                    |                                                          |                                                     |
| Dyspepsia                                | 2 (29)                                                                                                                             | 3 (75)                                                   | 4 (57)                                              |
| Dystonia                                 | 0 (0)                                                                                                                              | 2 (50)                                                   | 2 (29)                                              |
| Food craving                             | 0 (0)                                                                                                                              | 2 (50)                                                   | 2 (29)                                              |
| Gastritis                                | 0 (0)                                                                                                                              | 1 (25)                                                   | 1 (14)                                              |
| Hallucination, auditory                  | 0 (0)                                                                                                                              | 2 (50)                                                   | 2 (29)                                              |
| Intervertebral disc herniation           | 0 (0)                                                                                                                              | 1 (25)                                                   | 1 (14)                                              |
| Nightmare                                | 0 (0)                                                                                                                              | 2 (50)                                                   | 2 (29)                                              |
| Nocturia                                 | 0 (0)                                                                                                                              | 1 (25)                                                   | 1 (14)                                              |
| Paresthesia                              | 0 (0)                                                                                                                              | 2 (50)                                                   | 2 (29)                                              |
| Yawning                                  | 0 (0)                                                                                                                              | 1 (25)                                                   | 1 (14)                                              |

Abbreviation: AE, adverse events.

This table includes the five most common AEs (based on data in CSRs) and 10 randomly selected AEs. It shows the number of trials that reported results with sufficient information to calculate the between-arm effect and its precision in public and nonpublic sources. All AEs, including all serious AEs, are listed in [Appendix 4](#) (gabapentin) and [Appendix 5](#) (quetiapine).

Public sources: journal articles, conference abstracts, Food and Drug Administration reviews, trial registrations, and other short reports. Nonpublic sources: clinical study reports (CSRs), CSR-synopses, and individual participant data.

trials, respectively; most of them were reported only in nonpublic sources. In gabapentin and quetiapine trials, 56 of 72 (78%) and 39 of 46 (85%) serious AEs were *not* reported in public sources ([Table 3](#)). Public

sources described fewer serious AEs than nonpublic sources for gabapentin (mean  $\leq 1$  [1] public, 10 [7] nonpublic) and quetiapine (mean  $\leq 1$  [1] public, 16 [13] nonpublic).



**Table 3.** Description of adverse event results reported in public and nonpublic sources

|                                                                                                                                                                                          | Gabapentin for neuropathic pain |                   |                | Quetiapine for bipolar depression |                   |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------|----------------|-----------------------------------|-------------------|----------------|
|                                                                                                                                                                                          | Public sources                  | Nonpublic sources | All sources    | Public sources                    | Nonpublic sources | All sources    |
| Number of trials                                                                                                                                                                         | 20 (95%)                        | 6 (29%)           | 21             | 7 (100%)                          | 4 (57%)           | 7              |
| Trials reporting results for the number of participants who experienced one or more AEs with sufficient information to calculate the between-arm effect and its precision                | 5 <sup>a</sup> (25%)            | 6 (67%)           | 9 <sup>a</sup> | 3 <sup>a</sup> (43%)              | 4 (100%)          | 6 <sup>a</sup> |
| Trials reporting results for the number of participants who experienced one or more <i>serious</i> AEs with sufficient information to calculate the between-arm effect and its precision | 4 <sup>b</sup> (20%)            | 6 (67%)           | 9 <sup>c</sup> | 4 <sup>d</sup> (57%)              | 4 (100%)          | 5 <sup>a</sup> |
| Number of adverse events                                                                                                                                                                 |                                 |                   |                |                                   |                   |                |
| Unique nonsystematic AEs                                                                                                                                                                 | 68 (16%)                        | 391 (93%)         | 419            | 35 (7%)                           | 463 (98%)         | 471            |
| Nonsystematic AEs with sufficient information to calculate the between-arm effect and its precision                                                                                      | 62 (15%)                        | 384 (93%)         | 411            | 23 (5%)                           | 459 (99%)         | 462            |
| Serious nonsystematic AEs                                                                                                                                                                | 16 (22%)                        | 63 (88%)          | 72             | 7 (15%)                           | 46 (100%)         | 46             |
| Serious nonsystematic AEs with sufficient information to calculate the between-arm effect and its precision                                                                              | 16 (23%)                        | 60 (86%)          | 70             | 1 (3%)                            | 38 (100%)         | 38             |
| AE groups                                                                                                                                                                                | 2 (17%)                         | 12 (100%)         | 12             | 4 (15%)                           | 26 (100%)         | 26             |

Abbreviation: AE, adverse events.

<sup>a</sup> Additionally, one trial reported results with insufficient information to calculate the between-arm effect and its precision.

<sup>b</sup> Additionally, four trials reported results with insufficient information to calculate the between-arm effect and its precision.

<sup>c</sup> Additionally, three trials reported results with insufficient information to calculate the between-arm effect and its precision.

<sup>d</sup> Additionally, two trials reported results with insufficient information to calculate the between-arm effect and its precision.

Public sources: journal articles, conference abstracts, Food and Drug Administration reviews, trial registrations, and other short reports. Nonpublic sources: clinical study reports (CSRs), CSR-synopses, and individual participant data.

### 3.3. Public sources did not report AEs grouped by anatomic or physiological system

Where we could identify the system used to code and group AEs, most gabapentin trials used COSTART [26], and most quetiapine trials used MedDRA [25] (Table 1).

All CSRs for gabapentin (6 of 6, 100%) and quetiapine trials (2 of 2, 100%) reported AEs grouped by anatomic or physiological system; however, no corresponding public source for those trials reported AEs grouped by anatomic or physiological system. Of the trials reported only in public sources, 2 of 15 (13%) gabapentin and 0 of 3 (0%) quetiapine trials reported AEs by anatomic or physiological system.

When grouped by anatomic or physiological system, some AEs were relatively common among participants in all intervention arms. For example, “Body As A Whole” was common in gabapentin trials and “Psychiatric Disorders” was common in quetiapine trials. There were also clear differences between the drug and placebo arms (Appendices 2 and 3). Compared with placebo, gabapentin was associated with more AEs related to the nervous system (e.g., dizziness, amnesia, somnolence, tremor) and the digestive system (e.g., diarrhea, nausea, vomiting), whereas quetiapine was associated with more AEs related to the nervous system (e.g., convulsions, dizziness, migraine) and

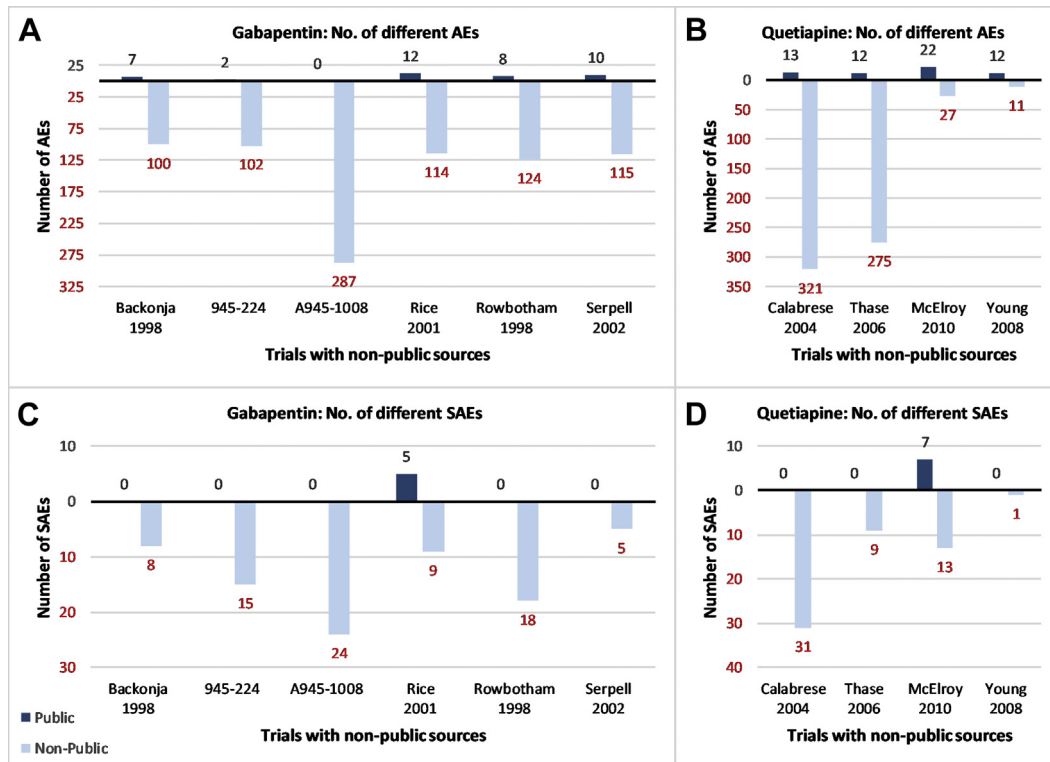
gastrointestinal disorders (e.g., abdominal pain, constipation, nausea). Quetiapine was also associated with more AEs related to eye disorders (e.g., diplopia, vision blurred, visual acuity reduced), yet neither specific eye disorders nor grouped eye disorders were described in public sources.

### 3.4. Public sources were insufficient for synthesis and decision-making

It would be impossible to synthesize information about AEs for all gabapentin or quetiapine trials in a meta-analysis. Using all available sources, a meta-analysis examining the risk of experiencing one or more AEs would include 9 of 21 (43%) gabapentin and 6 of 7 (86%) quetiapine trials (Appendix 4). Using only public sources, the same analysis would include 5 of 21 (24%) gabapentin and 3 of 7 (43%) quetiapine trials (Table 3). We found similar results when examining the number of participants who experienced one or more serious AEs (Table 3).

## 4. Discussion

In Part 2 of this study, we found that public sources of clinical trials reported little information about which AEs



**Fig. 1.** Fewer nonsystematic adverse events are reported in public sources compared with nonpublic sources: (A) Gabapentin: No. of different AEs in trials with non-public sources. (B) Quetiapine: No. of different AEs in trials with non-public sources. (C) Gabapentin: No. of different SAEs in trials with non-public sources. (D) No. of different SAEs in trials with non-public sources. Public sources: journal articles, conference abstracts, Food and Drug Administration reviews, trial registrations, and other short reports. Nonpublic sources: clinical study reports (CSRs), CSR-synopses, and individual participant data (IPD). This figure includes CSRs and IPD for all gabapentin trials and Calabrese 2004 and Thase 2006 (quetiapine trials). This figure includes only CSR-synopses for McElroy 2010 and Young 2008. AEs, adverse events. SAEs, serious adverse events. No. of AEs, number of AEs reported in each type of source. For example, if a source contained information about “dizziness,” “confusion,” and “somnolence,” the number of adverse events would be 3. No. of SAEs, number of serious AEs reported in each source.

participants experienced and the relative likelihood of experiencing those AEs between arms. It is especially concerning that most serious AEs were not described in public sources. Nonsystematic AEs were grouped for analysis in nonpublic sources, yet public sources almost never reported grouped AEs. For example, FDA prescribing information warns that cataracts have been associated with quetiapine in studies of dogs [42]; we found that eye disorders have been observed in human trials, but no public sources indicated that quetiapine has been associated with eye disorders in people.

All AEs that occur in clinical trials should be reported publicly to prevent reporting bias [43]. That is, systematic reviews and clinical guidelines that rely on public information about nonsystematic AEs might miss important AEs and include systematically biased information about other AEs. If public reports of clinical trials include only common AEs, and if public reports include only evidence that confirms certain AEs are related to interventions, then both rare AEs and disconfirming evidence will be overlooked by decision-makers. Although decision-makers might be overwhelmed [44] if information about hundreds of nonsystematic AEs were reported in journal articles and other

summaries of clinical trials (e.g., Appendices 4 and 5) [45,46], information about all nonsystematic AEs could be reported in public registries such as [www.ClinicalTrials.gov](http://www.ClinicalTrials.gov).

Most trials are not designed to compare individual nonsystematic AEs between arms. Grouping AEs by anatomic or physiological category can increase statistical power, and it can limit the number of comparisons and the potential for false positives in clinical trials [47]. Grouping might also allow investigators to hide certain AEs by combining important AEs with less important AEs. For example, “headache” and “tremor” are both “nervous system disorders” that can be important to patients; the latter might be more likely to be important and long-lasting. To prevent reporting bias, methods for grouping and analyzing nonsystematic AEs should be described in trial protocols and statistical analysis plans.

Because nonsystematic AEs are not collected using specific tools or instruments, nonsystematic AEs might be assessed inconsistently across trials or even within a trial by investigators who use different procedures to assess an outcome. Although it might be impossible to anticipate which AEs will occur in the first clinical trial of a new intervention, AEs could be identified for systematic collection based on

preclinical studies, trials of related interventions (e.g., drugs in the same class), observational studies, and studies of the same intervention for other indications [46]. As described in Part 1, systematic collection of anticipated and patient-important AEs would promote greater consistency across trials and across sites within individual trials [17].

This study was limited to two drugs and health conditions, and eligible trials were relatively short in duration. Additional research is needed to assess whether our findings apply to other interventions and longer trials in which more AEs and more serious AEs occur. Our findings were similar for both cases, and our findings are consistent with previous studies that compared public and nonpublic sources [8–11]; although the clinical implications of underreporting nonsystematic AEs vary across interventions and conditions, our methodological findings might be generalizable beyond these two drugs and conditions.

“Open science” and data sharing offer partial solutions to the problems identified in this study. In future studies, investigators could report all AEs in trial registries and through data sharing services such as Vivli (<https://vivli.org/>). Many commonly prescribed medications were evaluated and approved by regulators before trial registration and reporting requirements were commonplace [20,48]; nonetheless, it is important for stakeholders to know the AEs associated with interventions in current use, regardless of when those interventions were evaluated in clinical trials. Standards are needed for collecting, summarizing, sharing, and synthesizing AEs in clinical trials [17], including standards for sharing the results of “legacy” studies [49].

### CRedit authorship contribution statement

**Evan Mayo-Wilson:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Writing - original draft, Writing - review & editing. **Nicole Fusco:** Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing - review & editing, Writing - original draft. **Tianjing Li:** Conceptualization, Funding acquisition, Investigation, Methodology, Visualization, Writing - original draft, Writing - review & editing. **Hwanhee Hong:** Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing - review & editing. **Joseph K. Canner:** Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing - review & editing. **Kay Dickersin:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing - review & editing, Writing - original draft.

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### Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclinepi.2019.04.020>.

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